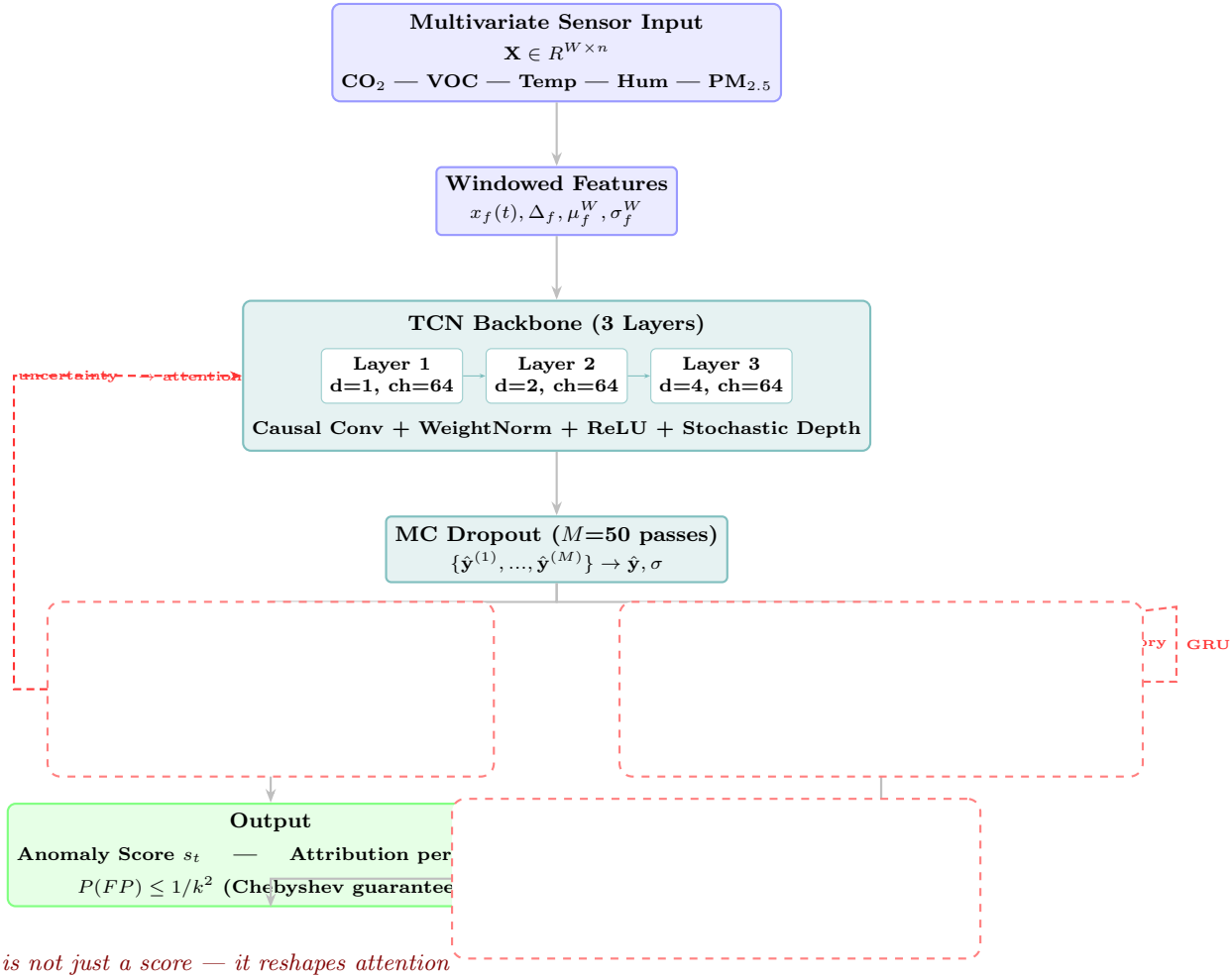
Uncertainty-Calibrated TCN with Adaptive Attention for Sensor Anomaly Detection



Key insight: *uncertainty is not just a score — it reshapes attention*

Component	ΔMSE	Cumul.
Base TCN	—	0.000327
+ MC Dropout	-5%	0.000311
+ AUAA (1)	-25%	0.000245
+ Full (1+2+3)	-40%	**0.000196**

innovation components